From Tweets to Trends

Predicting Stock Volume Using Social Sentiment

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1 Introduction

In the era of real-time communication, social media platforms like X (formerly Twitter) have become critical arenas for gauging market sentiment. This project explores how public discourse on X influences trading behavior, particularly focusing on NVIDIA's stock (NVDA). We developed an end-to-end data pipeline and machine learning system to forecast next-day trading volume based on tweet sentiment using Microsoft Azure's big data ecosystem.

Business Motivation

Financial firms increasingly rely on sentiment-driven analytics to anticipate market movements. Key applications of our system include:

- **Algorithmic Trading**: Sentiment forecasts support real-time, automated trading strategies.
- **Market Risk Management**: Early identification of panic signals enables timely hedging.
- **Corporate Communication**: Companies like NVIDIA can monitor sentiment to tailor responses.
- **Influencer and Marketing Analysis**: Identifies individuals and topics driving market buzz.

Project Scope and Focus

We chose to model NVIDIA stock exclusively for higher precision. An event-centric rationale guided this choice: the launch of DeepSeek, initially seen as a competitor, stirred negative sentiment that shifted once investors recognized

its dependence on NVIDIA GPUs. This event revealed how real-time sentiment can affect stock behavior, especially volume.
We focused on predicting **trading volume** rather than price or return. Volume is more stable and interpretable, serving as a strong proxy for investor attention without the volatility of price predictions.

Data Architecture and Sources

Our architecture follows the **Medallion Architecture** (Bronze, Silver, Gold layers):

- **Bronze Layer**: Ingests raw data from RapidAPI (Twitter) and Yahoo Finance (NVDA stock).
- **Silver Layer**: Cleans and normalizes tweets and stock data for alignment.
- **Gold Layer**: Merges sentiment and market data, enriching it with engineered features for modeling and visualization.

Azure services such as **Data Factory, Synapse Notebooks, and Function Apps** support both **batch** and **streaming** pipelines, making our system robust and real-time capable.

Data Pipelines

Tweet Pipeline: Using Azure Data Factory, we queried tweets with `#$NVDA`, handled pagination with custom logic, and stored them in Blob Storage. Features included sentiment (via VADER), engagement ratios, user credibility, and categorical flags (e.g., viral, influencer).
Stock Pipeline: Leveraged `yfinance` to fetch daily NVDA stock data. Preprocessed data aligned with tweet records by date for integration in the Gold layer.

Exploratory Insights

Exploratory data analysis revealed:

- Volume spikes align with sentiment swings (e.g., DeepSeek event).
- Verified users generate higher engagement, supporting credibility weighting.
- Favorites are more reliable than replies/retweets for sentiment interpretation.
- Co-mentions with $TSLA, $AMD, and $MSFT suggest thematic linkages.

Machine Learning Pipeline

We trained a **Random Forest Regressor** using **Spark MLlib** to predict next-day volume. Features included sentiment scores, engagement metrics, user attributes, and time-related flags. A feedback loop retrains the model when error exceeds 1%.

Daily Prediction and Feedback Loop

- **5:00 AM**: Tweets and stock data collected
- **6:00 AM**: Model runs prediction
- **6:30 PM**: Actual volume retrieved, error computed
- Retraining triggered if relative error $> 1\%$

Visualization and Business Intelligence

Gold-layer aggregations support a **Power BI Dashboard**, featuring:

- Sentiment vs. Volume trends
- Influencer engagement tracking
- Viral tweet logs and word clouds

Data is processed into fact/dimension tables and refreshed daily via Synapse pipelines.

Optimizations and Challenges

- **Data Skew**: Mitigated using log transformations
- **API Pagination**: Handled with cursor tracking logic
- **Cost Control**: Automated SQL pool orchestration
- **Feedback Tuning**: Threshold set at 1% error

Conclusion and Future Work

We built a scalable, cloud-native system that predicts trading volume based on public sentiment. Future work includes:

- Incorporating weekend tweets for Monday forecasting
- Modeling tweet volume as a predictive signal
- Extending to multi-stock analysis
- Using advanced NLP for deeper sentiment understanding

For a more detailed explanation of our architecture, implementation, and results, please refer to the full project report.